MUHAMMAD WASSAY SHAHID

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EDUCATION

McMaster University Sep 2022 – Apr 2027 (Expected)
Bachelor of Computer Science, Honours Co-op *Hamilton, ON*
Coursework: Data Structures & Algorithms, Operating Systems, Networks, Databases, Machine Learning.

EXPERIENCE

Software Engineer Co-op Sep 2025 – Aug 2026
Ontario Ministry of Public and Business Service Delivery *St. Catharines, ON*

- Developed responsive web applications and self-service portals for ministry staff and external vendors using Power Pages, Power Apps, JavaScript, HTML, and CSS.
- Built reusable custom front-end components in JavaScript, HTML, and CSS to extend the platform's built-in UI.
- Automated end-to-end service request workflows with Power Automate, connecting web forms to back-end data and cutting manual processing by **30%**.
- Integrated apps with Azure and Microsoft 365 for authentication, data storage, and cross-system workflows.
- Ensured WCAG/AODA accessibility compliance through functional, regression, and accessibility testing, fixing defects and shipping iterative UX improvements.

AI/ML Intern May 2025 – Jul 2025
ArhamSoft (Pvt) Ltd. *Lahore, Pakistan*

- Designed and benchmarked **3** CNN architectures in PyTorch on CIFAR-10 (**60,000** images, **10** classes): a custom 7-layer CNN, a Deep Layer Aggregation network, and ResNet18.
- Reached **94.42%** test accuracy with ResNet18, up from **91.88%** for the custom CNN, using residual connections, batch normalization, and SGD with cosine annealing over **100** epochs.
- Reduced overfitting with data augmentation (random crops, horizontal flips, normalization), dropout, and weight decay.
- Deployed a trained model as a Streamlit web app that classifies uploaded images, showing top predictions and flagging low-confidence results.
- Wrote a model comparison report on accuracy vs. compute trade-offs to justify the final architecture choice.

PROJECTS

Syllabus Agent – AI Deadline Tracker github.com/WassayShahid/syllabus-agent

- Built a full-stack app (Python, React, TypeScript) that turns syllabus PDFs into a deadline database and timeline, deployed serverlessly on AWS Lambda, Step Functions, API Gateway, S3, DynamoDB, and Cognito via AWS CDK.
- Extracted deadlines with Claude on Amazon Bedrock using a strict JSON tool schema plus code validation, scoring **100%** (**33/33** dates and weights) on a hand-labelled eval of **5** real course outlines.
- Built a tool-using AI agent with RAG (Titan embeddings) that answers questions with cited syllabus lines.
- Added daily email reminders with EventBridge and Amazon SES; shipped with **61** unit tests and GitHub Actions CI/CD.

Moodify – Mood-Based Music Recommender github.com/WassayShahid/Moodify

- Built a Flask and JavaScript web app that reads mood from facial expressions in-browser with face-api.js and plays matching tracks from a Spotify playlist.
- Sorted up to **300** tracks into **6** moods via weighted Last.fm tags fetched concurrently and cached in SQLite; Spotify OAuth saves picks as a playlist.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, HTML, CSS, Bash
AI/ML: Amazon Bedrock, LLM Agents, RAG, Embeddings, PyTorch, TensorFlow, OpenCV, pandas, NumPy
Cloud: AWS (Lambda, Step Functions, API Gateway, S3, DynamoDB, Cognito, SES, CDK, CloudWatch), Azure
Tools: React, Vite, Flask, SQLite, REST APIs, Git, GitHub Actions (CI/CD), pytest, JUnit, Linux, Agile